

## Scalability & Future Plan

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|----------------------|----------------------------------------------------|
| <b>Date</b>          |                                                    |
| <b>Team ID</b>       | xxxxxx                                             |
| <b>Project Name</b>  | EduGenie: Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 1 Mark                                             |

### Scalability & Future Plan:

How EduGenie can scale to more users and features, and the roadmap beyond the current version.

#### Current System Limitations:

| S.No | Limitation                                               | Impact                                         | Priority to Address (High / Medium / Low) |
|------|----------------------------------------------------------|------------------------------------------------|-------------------------------------------|
| 1    | Depends on Gemini API availability and rate limits       | Requests may fail or slow down under heavy use | High                                      |
| 2    | No user accounts or saved history                        | Learners can't track progress over time        | Medium                                    |
| 3    | Runs locally (not hosted); English-only, text-only input | Limited reach; no voice/image/PDF input        | Medium                                    |

#### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                                         | Proposed Upgrade / Solution                                                                 |
|------|--------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1    | User Load          | Single Uvicorn worker; 50 concurrent users with 0% errors (server overhead ~12-58 ms) | Containerize (Docker) and deploy to Google Cloud Run with auto-scaling and multiple workers |
| 2    | Data Storage       | Stateless - no database                                                               | Add Firestore / Cloud SQL for user profiles, quiz history and progress                      |
| 3    | Performance        | Every request calls Gemini                                                            | Cache frequent answers and learning paths, stream responses, use a paid Gemini tier         |
| 4    | Security           | API key in .env; no authentication                                                    | Google Sign-In / OAuth, Secret Manager for keys, rate limiting per user, HTTPS              |

#### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                                                     | Target Timeline | Expected Impact                                |
|---------|-------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------|
| Phase 2 | Cloud deployment, user accounts, saved quiz history, progress dashboard                                           |                 | Anytime access and visible learning progress   |
| Phase 3 | Voice interaction, multilingual support, image/PDF input for doubt solving                                        |                 | Accessibility for more learners and languages  |
| Phase 4 | Mobile app, gamification (badges, streaks), teacher/parent dashboards, LMS integration (Moodle, Google Classroom) |                 | Higher engagement and adoption by institutions |